2022 SCIENCE OLYMPIAD NATIONAL INVITATIONAL

JANUARY 8, 2022

Dynamic Planet C

Directions:

- You will have 50 minutes to complete the entire exam.

- You may use a binder, but this test is not open internet. It is recommended that you have a calculator.

- The point value of "select all that apply" questions may or may not be equal to he number of correct answers.

FROM THE HOSTS OF THE 2022 NATIONAL TOURNAMENT

SOUTHERN CALIFORNIA SCIENCE OLYMPIAD

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1. (1 point) Assuming no other fluxes exist, when evaporation exceeds precipitation, the water level of a lake will:

   A. rise

   B. fall

   C. remain the same

2. (1 point) Which of the streams below has a higher competence?

   Stream 1



   Stream 2:

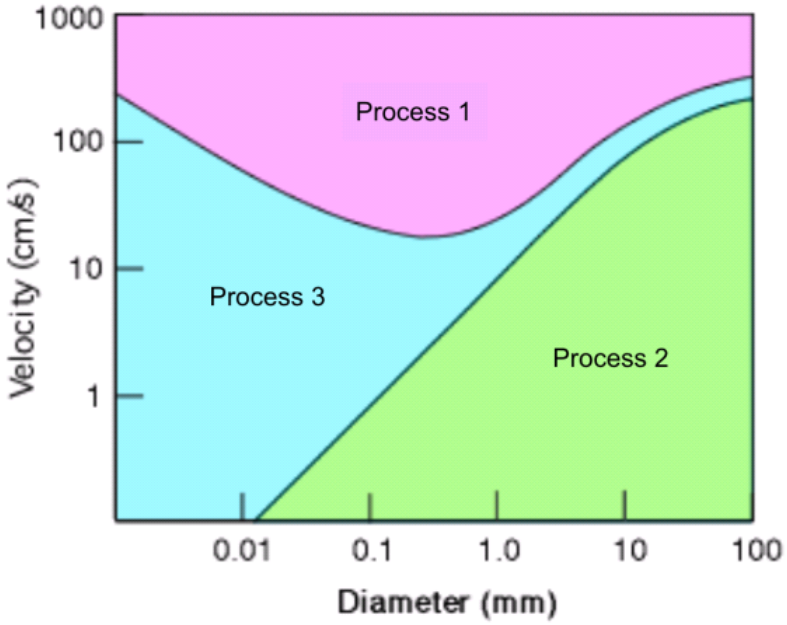


   A. Stream 1

   B. Stream 2

   C. Cannot be determined without additional information

3. (1 point) Which of the following properly labels the processes in the diagram below?



A. 1 - deposition, 2 - erosion, 3 - transportation

B. 1 - deposition, 2 - transportation, 3 - erosion

C. 1 - erosion, 2 - deposition, 3 - transportation

D. 1 - erosion, 2 - transportation, 3 - deposition

E. 1 - transportation, 2 - erosion, 3 - deposition

F. 1 - transportation, 2 - deposition, 3 - erosion

4. (2 points) Which of the following would decrease the permeability of some sediment?

√ **Decreasing the number of conduits**

◯ Widening the existing conduits

◯ Straightening non-straight conduits

◯ Changing the particle size from clay to gravel

√ **Changing the particle size from gravel to clay**

5. (1 point) Which of the following is a nonpoint source pollutant?

A. Sewage treatment plant

B. Car tailpipes

C. Factory

D. City storm drain

6. (3 points) Which of the following would serve as aquitards?

◯ Gravel

√ **Shale**

√ **Evaporites**

◯ Limestone

◯ Poorly-cemented sandstone

√ **Well-cemented sandstone**

7. (2 points) Which of the following is true regarding saltwater intrusion?

◯ Saltwater intrusion often enhances groundwater quality

√ **Saltwater intrusion is likely to occur in coastal aquifers**

◯ Saltwater intrusion can be mitigated by thoroughly pumping out groundwater

◯ Sea level rise lessens the effects of saltwater intrusion

√ **Storm surges amplify the effects of saltwater intrusion**

8. (1 point) Which of the following drainage patterns is likely to form in a region of alternating bands of siltstone and gneiss?

A. Dendritic

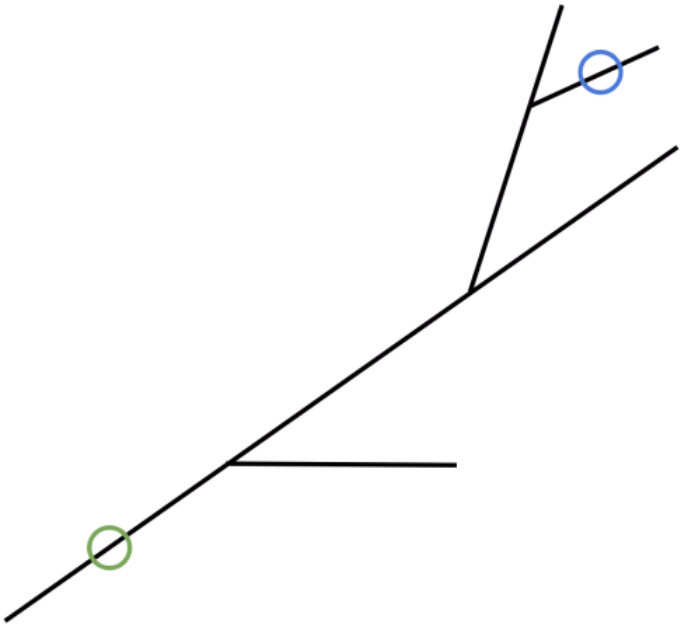
B. Parallel

C. Rectangular

D. Trellis

E. Deranged

For the next six questions, use the diagram below:



9. (1 point) What is the classic stream order of the portion of the stream with the green circle?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

10. (1 point) What is the classic stream order of the portion of the stream with the blue circle?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

11. (1 point) What is the Strahler stream order of the portion of the stream with the green circle?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

12. (1 point) What is the Strahler stream order of the portion of the stream with the blue circle?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

13. (1 point) What is the Shreve stream order of the portion of the stream with the green circle?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

14. (1 point) What is the Shreve stream order of the portion of the stream with the blue circle?

A. 1

B. 2

C. 3

D. 4

E. 5

F. 6

15. (1 point) In January, which of the following layers is the warmest?

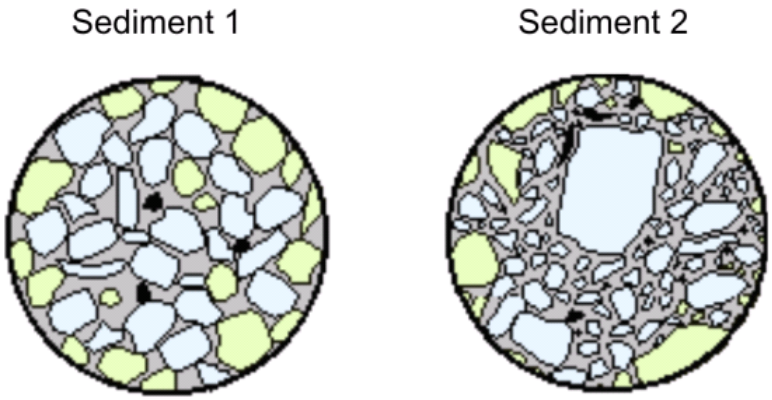
A. epilimnion

B. metalimnion

C. hypolimnion

D. none of the above

Use the diagram below to answer the following four questions. Gray represents pore space.



16. (1 point) Which sediment depiction would be found in an alluvial fan?

A. Sediment 1

B. Sediment 2

17. (1 point) Which sediment depiction would be found in a river bed?

A. Sediment 1

B. Sediment 2

18. (1 point) Which sediment depiction would be found in glacial till?

A. Sediment 1

B. Sediment 2

19. (1 point) Which sediment would have a higher hydraulic conductivity?

A. Sediment 1

B. Sediment 2

20. (2 points) Which of the conditions would be expected in a region exhibiting karst topography?

√ **Calcareous rock composition**

◯ Arid climate

◯ Hydrothermal

√ **Low groundwater pH**

√ **Warm temperatures**

21. (1 point) In which of the following rock types would pseudokarst sinkholes not be found?

    A. Limestone

    B. Sandstone

    C. Quartzite

    D. Gypsum

    E. Pseudokarst sinkholes form in all of the above rock types

Use the image below to answer the following two questions.



22. (1 point) What feature is shown in the image?

    A. Alluvial fan

    B. Artificial levees

    C. Braided streams

    D. Delta

    E. Oxbow lake

23. (1 point) Which of the following statements is true regarding this type of feature?

    A. These features tend to coalesce with the main stream channel over time

    B. These features tend to fill up with sediment over time

    C. These features tend to get deeper over time due to water erosion

    D. These features tend to form from gradually rising water table levels

24. (1 point) Which of the following most accurately describes the direction of groundwater flow?

    A. Groundwater generally flows parallel to the contour lines of a topographic map

    B. Groundwater generally flows uphill following rainstorms

    C. Groundwater generally flows parallel to the water table contours

    D. Groundwater generally flows perpendicular to the water table contours

25. (1 point) Which of the following would likely form a cone of depression?

    A. Channelization

    B. River damming

    C. Well pumping

    D. Aquitard formation

E. Groundwater recharge

26. (1 point) Which of the following would not be a significant source of nitrate pollution?

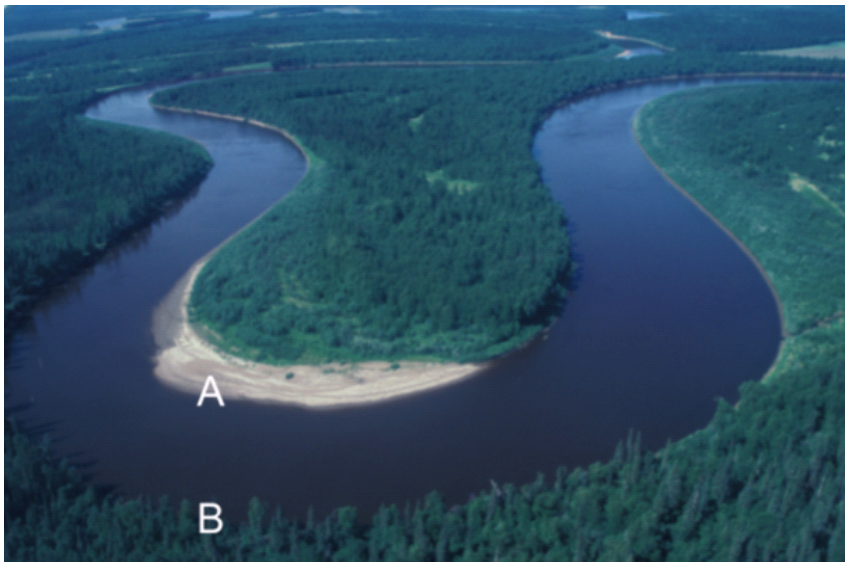
A. Acid rain

B. Fertilizer runoff

C. Undertreated sewage leaks

D. Contamination by livestock waste

Use the diagram below to answer the following two questions.



27. (1 point) The flow velocity in the stream near location A is $_{t he flow velocity near location B.}$

A. higher than

B. lower than

C. equal to

28. (2 points) Which of the following statements regarding this stream are correct?

◯ Location A is known as the cut bank

◯ Location B is known as the point bar

◯ Erosion dominates at location A

√ **Erosion dominates at location B**

√ **Deposition dominates at location A**

◯ Deposition dominates at location B

29. (1 point) Which of the following stages results from the longest duration of stream erosion?

A. Deeply cut V-shaped valley

B. Deeply cut U-shaped valley

C. A series of waterfalls

D. Flattened topography near sea level

E. Radial drainage pattern

30. (1 point) Which of the following results when the potentiometric surface is above the ground?

    A. Braided stream

    B. Meandering stream

    C. Effluent stream

    D. Influent stream

    E. Flowing artesian well

    F. Non-flowing artesian well

For the following 12 questions, identify the term corresponding to the description. Each term is exactly one word (if the description is describing a braided stream, only answer with "braided").

31. (1 point) The zone of permeable substrate around a permanent stream in which water flows and nutrients are cycled.

    hyporheic

32. (1 point) A water table or aquifer that is trapped by a localized aquitard higher than the regional water table.

    perched

33. (1 point) A nickpoint which commonly forms when an eroding river transitions from flowing over harder rock to softer rock.

    waterfall

34. (1 point) A stream that remains on its original course despite changes in the underlying surface.

    antecedent

35. (1 point) The type of porosity that takes into account the new pore space formed after the formation of the original rock.

    secondary

36. (1 point) The maximum amount of solid load that can be carried by a stream.

    capacity

37. (1 point) A stream that stems off of and flows away from the main stream.

    distributary

38. (1 point) Volcanic lakes such as Crater Lake can form in this type of depression that is formed following an eruption.

    caldera

39. (1 point) The line of lowest elevation in a stream or river valley.

    thalweg

40. (1 point) The Spanish name for an ephemeral stream in an arid region.

    arroyo

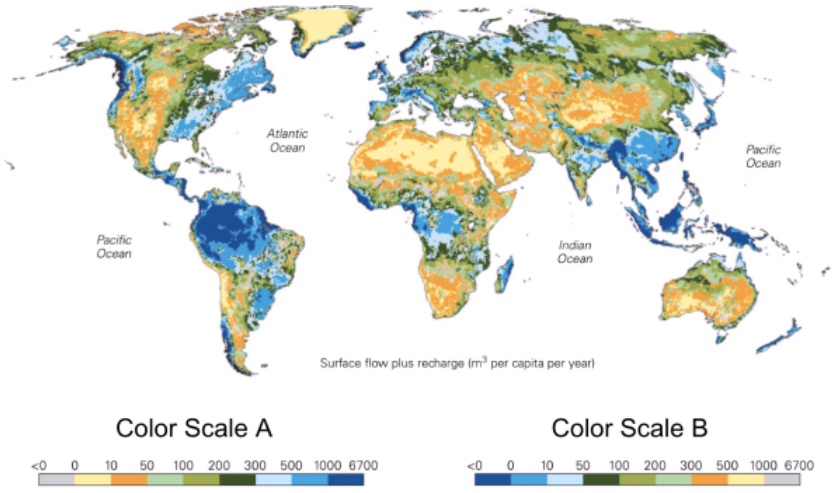
41. (1 point) The feature that separate neighboring drainage basins.

    divide

42. (1 point) This type of erosion causes the stream to lengthen in the direction opposite to its flow.

    headward

43. (4 points) This map plots surface flow plus recharge measured in cubic meters per capita per year.



a) Is Color Scale A or Color Scale B correct for this map? Briefly justify your choice.

b) What does a high or low value on the map represent in terms of the availability of water in an area?

Solution:

The following questions will use the topographic map of the USGS Wolf Mountain, Montana Quadrangle, which can be viewed by clicking this link: https://scilympiad.com/Data/turs/10RG/tests/00098G/USGSTopoMap-compressed-compressed-min.pdf

44. (2 points) What is the contour interval for the map?

Solution: 40 feet

45. (2 points) What direction does the Grey Blanket Creek flow in?

Solution: West

46. (2 points) What is the name for the location where streams join together?

Solution: confluence

47. (2 points) What type of drainage pattern is around Grey Blanket Creek?

Solution: Dendritic

48. (2 points) What type of stream is Grey Blanket Creek?

A. Anastamosing

B. Braided

C. Intermittent

D. Perennial

E. None of the above

49. (2 points) Among all the tributaries of Grey Blanket Creek, what is the elevation of the highest headwater?

Solution: 4840 feet

50. (2 points) Identify the elevation of any nick points that lie along Grey Blanket Creek.

Solution: none

51. (3 points) What is the highest elevation that is within the Grey Blanket Creek watershed?

Solution: 5160 feet. (2 pts if between 5120 and 5200 feet).

52. (6 points) A theoretical open stream channel has a rectangular cross-section that is 10 meters wide. If the discharge of this stream is 8 m3/s, the Manning's roughness coefficient is 0.02, and the channel slope is 1/1500 m/m, calculate the flow depth of the stream. Show all calculations.

Solution:

53. (6 points) A hydrogeologist makes measurements using two wells, Well A and Well B, that are 1 kilometer apart horizontally. For Well A, the top of the well is at an elevation of 808.8 meters and the water table is 16.1 meters from the top of the well. For Well B, the top of the well is at an elevation of 799.9 meters and the water table is 22.2 meters from the top of the well. If the hydraulic conductivity is 10 meters per day and the porosity of the rock is 0.25, calculate how many days it will take for water to flow from Well A to Well B. Show all work.

Solution:

54. (4 points) Briefly describe the difference between dimictic, monomictic, polymictic, and meromictic lakes.

Solution:

55. (3 points) Briefly describe three geologic settings in which groundwater springs form

Solution:

56. (2 points) Briefly describe the difference between bogs and marshes.

Solution:

57. (2 points) Briefly describe the difference between ephemeral and intermittent streams.

Solution:

58. (2 points) Briefly describe the difference between anastomosing and braided streams.

Solution:

59. (2 points) Briefly describe the difference between hydraulic and hydrostatic pingos.

Solution:

60. (3 points) Briefly describe the difference between the epilittoral, littoral and profundal zones of a lake

Solution:

61. (2 points) Briefly describe the difference between endorheic and exorheic basins.

Solution:

62. (3 points) Briefly describe a feedback loop involving permafrost and identify whether it is positive or negative.

Solution:

63. (2 points) Briefly describe the process of stream piracy.

Solution:

64. (3 points) Briefly describe a drawback of flood-control dams and a less consequential alternative to flood control.

Solution:

Conclusion: